

Part 1: S5

Use the simplified rules for **S5** to build tableaux for each of these claims. Say which of them are valid, i.e., are theorems of **S5**.

1. $\Box(\Box p \rightarrow (q \vee r)) \vee \Box(\Box q \rightarrow p)$ in **S5**

1.	$\mathbb{F} 1, \Box(\Box p \rightarrow (q \vee r)) \vee \Box(\Box q \rightarrow p)$ ✓	Assumption
2.	$\mathbb{F} 1, \Box(\Box p \rightarrow (q \vee r))$ ✓	$\vee\mathbb{F} 1$
3.	$\mathbb{F} 1, \Box(\Box q \rightarrow p)$ ✓	$\vee\mathbb{F} 1$
4.	$\mathbb{F} 2, \Box p \rightarrow (q \vee r)$ ✓	$\Box\mathbb{F} 2$
5.	$\mathbb{T} 2, \Box p$	$\rightarrow\mathbb{F} 4$
6.	$\mathbb{F} 2, q \vee r$ ✓	$\rightarrow\mathbb{F} 4$
7.	$\mathbb{F} 2, q$	$\vee\mathbb{F} 6$
8.	$\mathbb{F} 2, r$	$\vee\mathbb{F} 6$
9.	$\mathbb{F} 3, \Box q \rightarrow p$	$\Box\mathbb{F} 3$
10.	$\mathbb{T} 3, \Box q$	$\rightarrow\mathbb{F} 9$
11.	$\mathbb{F} 3, p$	$\rightarrow\mathbb{F} 9$
12.	$\mathbb{T} 3, p$	$\Box\mathbb{T} 5$
	⊗	

The only branch closes, so this is **valid** in **S5**.

$\Diamond\Box(p \rightarrow q) \rightarrow (\Box p \rightarrow \Diamond q)$ in **S5**

1.	$\mathbb{F} 1, \Diamond\Box(p \rightarrow q) \rightarrow (\Box p \rightarrow \Diamond q)$ ✓	Assumption
2.	$\mathbb{T} 1, \Diamond\Box(p \rightarrow q)$ ✓	$\rightarrow\mathbb{F} 1$
3.	$\mathbb{F} 1, \Box p \rightarrow \Diamond q$ ✓	$\rightarrow\mathbb{F} 1$
4.	$\mathbb{T} 1, \Box p$	$\rightarrow\mathbb{F} 3$
5.	$\mathbb{F} 1, \Diamond q$	$\rightarrow\mathbb{F} 3$
6.	$\mathbb{T} 2, \Box(p \rightarrow q)$	$\Diamond\mathbb{T} 2$
7.	$\mathbb{T} 2, p \rightarrow q$ ✓	$\Box\mathbb{T} 2$
8.	$\mathbb{T} 2, p$	$\Box\mathbb{T} 4$
9.	$\mathbb{F} 2, q$	$\Diamond\mathbb{F} 5$
10.	$\begin{array}{cc} \mathbb{F} 2, p & \mathbb{T} 2, q \end{array}$	$\rightarrow\mathbb{T} 7$
	⊗ ⊗	

Both branches close, so this is **valid** in **S5**.

$\Box(p \vee q) \rightarrow (\Diamond(\neg q \vee r) \rightarrow \Diamond(p \vee r))$ in **S5**

1.	$\mathbb{F} 1, \Box(p \vee q) \rightarrow (\Diamond(\neg q \vee r) \rightarrow \Diamond(p \vee r))$ ✓	Assumption
2.	$\mathbb{T} 1, \Box(p \vee q)$	$\rightarrow \mathbb{F} 1$
3.	$\mathbb{F} 1, \Diamond(\neg q \vee r) \rightarrow \Diamond(p \vee r)$ ✓	$\rightarrow \mathbb{F} 1$
4.	$\mathbb{T} 1, \Diamond(\neg q \vee r)$ ✓	$\rightarrow \mathbb{F} 3$
5.	$\mathbb{F} 1, \Diamond(p \vee r)$	$\rightarrow \mathbb{F} 3$
6.	$\mathbb{T} 2, \neg q \vee r$ ✓	$\Diamond \mathbb{T} 4$
7.	$\mathbb{T} 2, p \vee q$ ✓	$\Box \mathbb{T} 1$
8.	$\mathbb{F} 2, p \vee r$ ✓	$\Diamond \mathbb{F} 5$
9.	$\mathbb{F} 2, p$	$\vee \mathbb{F} 8$
10.	$\mathbb{F} 2, r$	$\vee \mathbb{F} 8$
└──┬──		
11.	$\mathbb{T} 2, p$ $\mathbb{T} 2, q$	$\vee \mathbb{T} 7$
	⊗	
	└──┬──	
12.	$\mathbb{T} 2, \neg q$ ✓	$\vee \mathbb{T} 6$
13.	$\mathbb{F} 2, q$ $\mathbb{T} 2, r$	$\neg \mathbb{T} 12$
	⊗	

All branches close, so this is **valid** in **S5**.

$(\neg \Diamond p \vee \neg \Diamond \Diamond p) \rightarrow \neg \Diamond \Diamond \Diamond p$ in **S5**

1.	$\mathbb{F} 1, (\neg \Diamond p \vee \neg \Diamond \Diamond p) \rightarrow \neg \Diamond \Diamond \Diamond p$ ✓	Assumption
2.	$\mathbb{T} 1, \neg \Diamond p \vee \neg \Diamond \Diamond p$ ✓	$\rightarrow \mathbb{F} 1$
3.	$\mathbb{F} 1, \neg \Diamond \Diamond \Diamond p$ ✓	$\rightarrow \mathbb{F} 1$
4.	$\mathbb{T} 1, \Diamond \Diamond \Diamond p$ ✓	$\neg \mathbb{F} 3$
5.	$\mathbb{T} 2, \Diamond \Diamond p$ ✓	$\Diamond \mathbb{T} 4$
6.	$\mathbb{T} 3, \Diamond p$ ✓	$\Diamond \mathbb{T} 5$
7.	$\mathbb{T} 4, p$	$\Diamond \mathbb{T} 6$
└──┬──		
8.	$\mathbb{T} 1, \neg \Diamond p$ ✓	$\vee \mathbb{T} 2$
9.	$\mathbb{F} 1, \Diamond p$	$\neg \mathbb{T} 8$
10.	$\mathbb{F} 4, p$	$\Diamond \mathbb{F} 9$
11.	⊗	$\neg \mathbb{T} 8$
12.	$\mathbb{F} 1, \Diamond \Diamond p$	$\Diamond \mathbb{F} 11$
	$\mathbb{F} 3, \Diamond p$	
	⊗	

Both branches close, so this is **valid** in **S5**. There are many ways to close this one; this is the fastest I found, but it's definitely not the only way.